Low-Rank Native Query–Key Routing: Compact Memory Indexes without K/V Compression

Jorge Simão
EInnovator R&D – Center for the Sciences of Computation, Cognition, and Complexity

August 2026

Abstract

Progressive Retrieval Attention (PRA) can retain full native key/value (K/V) memory outside the active attention working set, but it still needs a compact persistent index to decide which chunks to materialize. One mean native key per chunk is cheap, yet averaging can erase a sparse key that is strongly aligned with a future query. We therefore learn discriminative low-rank projections of frozen native queries and keys, cache projected key rows as an address index, and rank chunks in the resulting interaction space. The objective preserves evidence-ranking utility rather than reconstructing the full native attention response. The index is used only for selection; chosen chunks contribute their unchanged original native K/V.

On 64 fresh QASPER identities, rank-16 routing raises four-chunk evidence recall from 0.117 for a full-width mean key to 0.258 (paired +0.141, 95% interval $[0.080, 0.210]$). The same architecture transfers to 2WikiMultiHopQA: the projection trained on prior Hotpot/QASPER validation identities gains 0.033 over mean without target supervision, and dataset-specific retraining raises recall from 0.0977 to 0.1903. A rank-8 index with eight static centroids retains 84.9% of that 2WikiMultiHopQA gain at approximately 256 FP32 routing-index bytes per chunk, excluding masks. A SmolLM2/Llama-family replication preserves the QASPER effect. The channel is not universal: BM25 dominates on HotpotQA, lexical routing remains stronger on MuSiQue, and validation-selected static lexical–low-rank fusion regresses on both new cohorts. Native-K/V causal controls further show that informed routing beats shuffled and bottom-ranked memory on QASPER while the frozen late-layer consumer still underperforms no memory. Low-rank native-QK is therefore a compact routing channel, not a universal retriever, not a K/V compressor, and not a complete memory-use solution.

1 Introduction

Long-memory systems face two different storage problems. The first is how to retain source content or its native transformer K/V. The second is how to find the small subset worth making attention-visible for a current query. Treating these as one problem encourages an ambiguous use of "compression." A method may compress active K/V, replace context with summaries, evict tokens, or compress only the index that names intact memory. Those interventions have different failure modes and different accounting.

PRA separates routing from reading. A persistent routing representation ranks logical memory chunks; selected chunk identities then resolve to their original native K/V. This preserves pretrained values and bounds the active working set, but makes routing quality consequential. A common baseline stores one mean key per chunk:

$$\bar{k}_C = \frac{1}{T}\sum_{i=1}^{T} k_i. \tag{1}$$

The mean is deterministic and cacheable. It is also a severe bottleneck. If 31 token keys are unrelated to a later question and one key is strongly aligned, the useful direction can disappear in the average.

This paper asks whether query-dependent address information in native attention keys can be retained in a compact routing index without rewriting the memory that will later be read. We project frozen native keys and the current native query into a small learned interaction space:

$$z_q = W_q \bar{q}, \qquad z_i = W_k u_i, \qquad \ell_i = \frac{z_q^\top z_i}{\sqrt{r}}. \tag{2}$$

The z_i rows are cacheable. They select chunk identities; they are never substituted for native values or attention-visible keys. The core boundary is

$$\text{address-index compression} \neq \text{backing-K/V compression}. \tag{3}$$

The map $(q, K) \mapsto (W_q q, W_k K)$ is discriminative, not reconstructive. Its target is evidence ordering at the materialization boundary:

$$\boxed{\text{preserve evidence-ranking utility}} \quad \text{rather than} \quad \boxed{\text{minimize native-response reconstruction error}}. \tag{4}$$

Low teacher overlap and non-monotonic rank later support this interpretation; they do not make it a theorem about native attention geometry.

The strongest result is a dataset-specific retrieval channel. Low-rank native interaction substantially improves over a mean key on fresh QASPER and transfers to 2WikiMultiHopQA, including a smaller zero-shot gain. It remains weaker than lexical methods on HotpotQA and MuSiQue. A second boundary appears after selection: better-selected memory can be causally preferable to wrong memory while still harming a frozen consumer relative to no memory. Thus

$$\text{routing utility} \neq \text{memory-consumption utility}. \tag{5}$$

The contributions are:

1. a low-rank native query–key formulation that compresses a persistent routing index while preserving backing native K/V;
2. fresh QASPER evidence that rank-16 native interaction improves four-chunk recall from 0.117 to 0.258 over a full-width mean key;
3. cross-dataset evidence that the architecture transfers to 2WikiMultiHopQA, including zero-shot gain, while MuSiQue remains lexical;
4. a routing-index frontier in which rank-8/eight-centroid routing retains 84.9% of the 2WikiMultiHopQA rank-16 gain at approximately 256 FP32 routing-index bytes/chunk, excluding masks;
5. a SmolLM2/Llama-family QASPER replication with different native geometry and tokenizer;
6. native-K/V causal controls that separate improved routing from beneficial memory consumption.

The preregistered response-distillation and landmark-selector program that led to this mechanism did not pass its deployment gate. We retain that negative study in the appendices rather than making it the paper's organizing story.

2 Low-Rank Native-QK Routing

2.1 A discriminative projection with a narrow obligation

For a chunk C of T tokens, frozen pre-RoPE keys are

$$K_C \in \mathbb{R}^{T \times H_{kv} \times d_h}. \tag{6}$$

The full stream preserves token-level query interaction but is expensive as a universally resident routing structure. One mean key reduces T rows to one full-width row, but can erase sparse address information. Low-rank routing instead flattens each token's native key heads to $u_i \in \mathbb{R}^{H_{kv} d_h}$ and projects it with Equation 2. It preserves multiple projected rows while reducing the interaction width from $H_{kv} d_h$ to r.

For Qwen3-0.6B at the measured layer, $H_{kv} d_h = 1024$ and the flattened query width is 2048. We evaluate $r \in \{8, 16, 32\}$. The bilinear projection contains 24,576, 49,152, or 98,304 learned parameters; the backbone remains frozen. Training uses validation-only evidence ranking plus native-response distillation. Evidence ranking defines the operational target; response distillation is a regularizer, not a reconstruction contract. Low teacher overlap in the final models shows that the learned map is best interpreted as a discriminative projection of native attention geometry for memory addressing, not a faithful reconstruction of the full native QK response.

2.2 Persistent index and unchanged memory

At index construction, the system caches $Z_C = W_k U_C$. Online routing projects the current query once, computes low-rank dots, reduces token scores within each chunk, and returns chunk identities. Static variants replace 32 projected rows with four or eight deterministic centroids, medoids, or farthest-first representatives. This compresses two independent axes:

$$\text{interaction width: } 1024 \to r, \tag{7}$$

$$\text{routing rows: } 32 \to m. \tag{8}$$

Width compression maps $d_{\text{native}} = 1024$ to $r \in \{8, 16, 32\}$; multiplicity compression maps $T = 32$ projected token rows to $m \in \{4, 8\}$ representatives. Up to dtype and masks, routing-index residency therefore obeys

$$B_{\text{index}} \propto mr. \tag{9}$$

More representatives need not require more storage when their interaction width is compressed aggressively.

Figure 1 makes the ownership boundary explicit. The routing index stores neither native values nor the keys later supplied to attention. After selection, the existing PRA cache materializes the original token K/V for the chosen chunks.

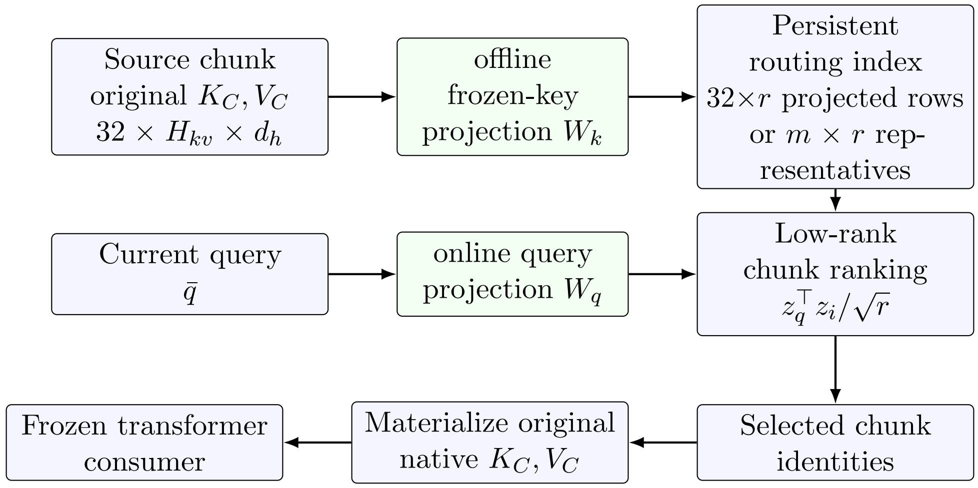

Figure 1: The low-rank representation is used only for address selection. Selected chunks contribute their unchanged original native K/V. Routing-index bytes and backing-K/V bytes are therefore separate quantities.

2.3 Index accounting

For FP32 routing rows, a 32-token Qwen chunk requires 131,072 bytes for the full 32×1024 native-key stream, 4,096 bytes for one full-width mean key, $128r$ bytes for all low-rank token rows, and $4mr$ bytes for static low-rank representatives. Thus rank-16 all-token routing uses 2,048 bytes/chunk, while rank-8/eight-centroid routing uses 256 bytes/chunk before masks. In particular, the latter replaces the 4,096-byte full-width mean index with eight projected representatives at 256 bytes/chunk. Backing native K/V, selected-K/V transfer, and active attention K/V are not included in these index figures.

Table 1: Representations used in the main experiments. Attention-visible state identifies what enters the frozen consumer after routing. Byte counts are FP32 routing-index residency per chunk and exclude masks and backing K/V.

Routing representation	Rows	Width	FP32 B/chunk	Attention-visible state
Full native-key stream	32	1024	131,072	unchanged selected native K/V
Native mean	1	1024	4,096	unchanged selected native K/V
Rank-16 projected tokens	32	16	2,048	unchanged selected native K/V
Rank-8/eight-centroid	8	8	256	unchanged selected native K/V

3 Experimental Setup

3.1 Model and chunk contract

The primary model is Qwen3-0.6B revision `c1899de2`, layer 27, with 16 query heads, eight native K/V heads, and head width 128. Bounded 256-token source calls are divided into 32-token routing chunks without local re-encoding. All methods request four chunks and materialize approximately 128 unchanged native K/V tokens. The routing layer uses pre-RoPE keys and the model's unchanged native query projection and normalization.

3.2 Cohorts

The central confirmation contains 64 fresh HotpotQA and 64 fresh QASPER identities, disjoint from the original 16+16 test matrix and validation identities [5, 6]. The cross-dataset extension uses 20 validation and 100 test identities each for 2WikiMultiHopQA and MuSiQue [7, 8]. Evidence labels come only from authored supporting sentences or paragraphs and exact tokenizer-aligned source spans. The SmolLM2 replication retrains the same projection contract on its own validation features and evaluates the fresh Hotpot/QASPER cohort.

3.3 Training regimes

Rank-8 and rank-16 projections train for five seeds on validation identities only. For 2WikiMultiHopQA and MuSiQue, *zero-shot* uses the projection trained on the prior Hotpot/QASPER validation cohort with no target-dataset supervision. *Dataset-specific retraining* uses only target validation identities. *Mixed training* uses a balanced four-dataset validation cohort. Neither rank nor objective is selected from the new test identities. The static lexical–low-rank fusion weight is validation-selected.

3.4 Metrics

Primary routing quality is physical evidence-chunk recall at $K = 4$. Contiguous evidence-group completion asks whether at least one chunk from every authored evidence region is selected. Paired 95% intervals use 5,000 identity bootstrap draws; learned scores ensemble five seeds unless noted. Systems results report routing-index bytes, cached component latency, and index construction separately. Native-K/V generation uses teacher-forced gold-token log-probability as primary because the frozen 0.6B model rarely emits exact answers in 12 greedy tokens.

4 QASPER and Hotpot: A Dataset-Specific Native Channel

The fresh confirmation is the primary result. On QASPER, rank-16 recall is 0.258 versus 0.117 for the native mean, a paired gain of 0.1407 with interval $[0.0795, 0.2099]$. Rank-8/eight-centroid reaches 0.206. Exact routing remains higher at 0.318, but rank 16 exceeds the validation-defined hybrid by 0.1149, interval $[0.0562, 0.1796]$. Native interaction therefore recovers a useful address signal that one mean erases; it does not make lexical controls obsolete.

The same mechanism does not help HotpotQA. Rank 16 reaches 0.120 versus 0.125 for mean, while BM25 reaches 0.386. The evidence form matters: Hotpot's entity and relation cues are well served by token-native lexical retrieval, whereas QASPER's research-paper questions expose a native interaction signal not captured by one mean.

Table 2: Fresh 64+64 confirmation at four selected chunks. Learned rows ensemble five validation-trained seeds.

Method	HotpotQA	QASPER
Native mean	0.125	0.117
BM25	**0.386**	0.065
Exact	0.282	**0.318**
Hybrid	0.224	0.143
Rank 16, all projected tokens	0.120	0.258
Rank 8, eight centroids	0.129	0.206
Evidence oracle	0.814	0.731

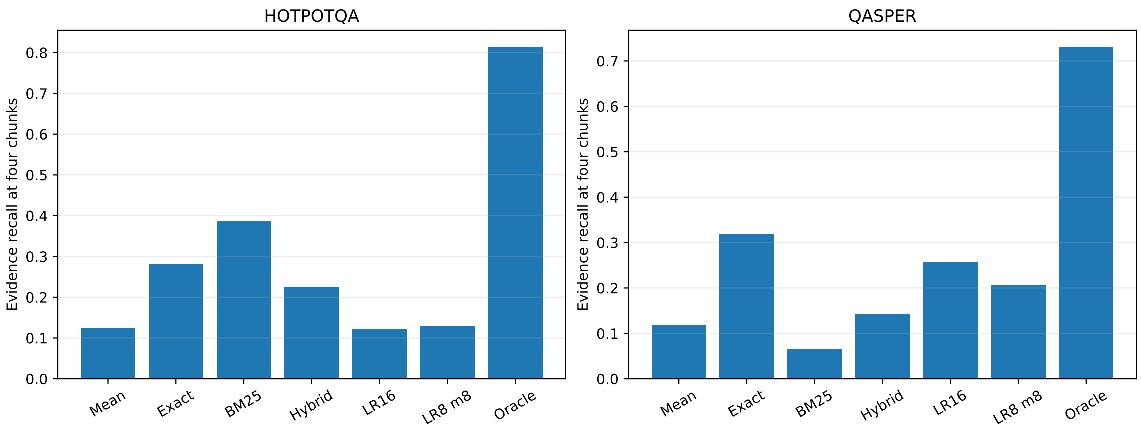

Figure 2: Fresh-cohort routing confirmation. Low-rank native interaction transfers within QASPER; HotpotQA remains lexical.

The confirmation was frozen after an earlier 16+16 diagnostic in which all-token rank 16 reached 0.254 QASPER recall and 0.121 HotpotQA recall. The fresh effect is therefore not a test-matrix reselection. Five diagnostic seed means ranged from 0.194 to 0.303 on QASPER, motivating seed ensembling and identity-paired intervals rather than a single-checkpoint claim.

5 Cross-Dataset Transfer: 2Wiki versus MuSiQue

The unchanged architecture transfers to a second positive dataset, but not to every multi-hop benchmark. On 2WikiMultiHopQA, mean recall is 0.0977. The rank-16 projection trained on the prior Hotpot/QASPER validation cohort reaches 0.1305 without 2WikiMultiHopQA supervision, a zero-shot gain of 0.0328 with interval $[0.0025, 0.0638]$. Dataset-specific retraining reaches 0.1903, gain 0.0926 with interval $[0.0582, 0.1289]$; balanced training reaches 0.1754. This separates architecture transfer from weight transfer: a useful geometry survives zero-shot, and the same compact family adapts more strongly with new validation supervision.

Exact routing remains the strongest 2WikiMultiHopQA channel at 0.282. On MuSiQue, zero-shot, retrained, and mixed rank-16 effects over mean are -0.0010, $+0.0043$, and $+0.0051$, with every interval crossing zero. Validation selects BM25, which reaches 0.122; held-out exact reaches 0.130 as a test-unselected diagnostic. The scientific result is therefore

$$\text{architecture transfer} \neq \text{universal channel utility.} \tag{10}$$

Table 3: Fresh cross-dataset routing at $K = 4$. R is physical evidence-chunk recall; C is contiguous evidence-group completion. Learned rows ensemble five seeds. Bold marks the highest reported held-out value, not a replacement selected after test inspection.

	2WikiMultiHopQA		MuSiQue	
Method	R	C	R	C
Native mean	.098	.10	.067	.13
Exact	**.282**	.18	**.130**	**.24**
BM25	.267	**.24**	.122	.20
Rank 16, zero-shot	.130	.10	.066	.11
Rank 16, retrained	.190	.18	.071	.15
Rank 16, mixed	.175	.20	.072	.15
Rank 8, eight centroids	.176	.15	.068	.14
Lexical + rank 16	.228	.21	.094	.14

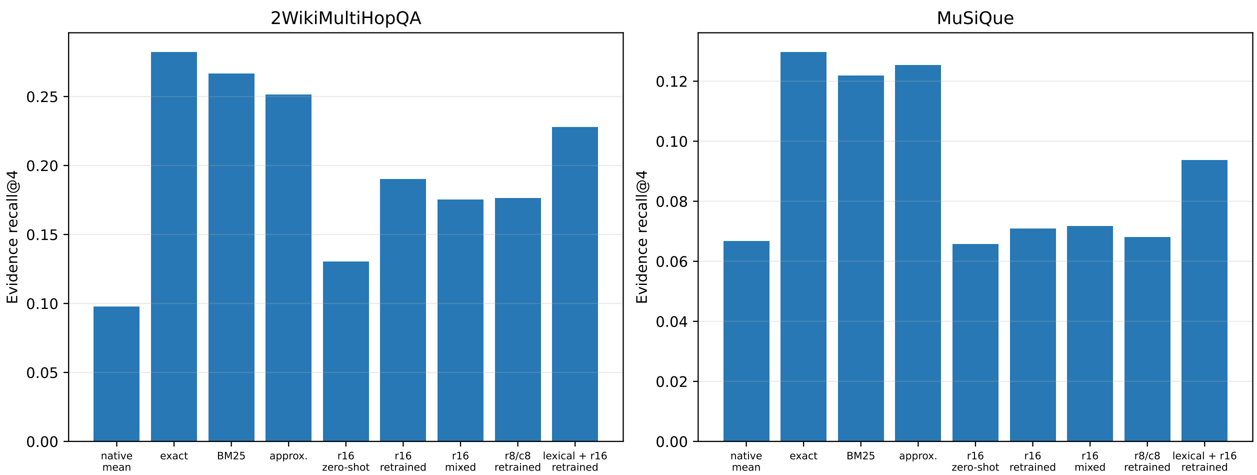

Figure 3: The low-rank channel transfers to 2WikiMultiHopQA but does not displace exact retrieval; MuSiQue remains lexical. The fusion row uses a validation-selected weight and illustrates failed calibration.

Static fusion fails despite genuine complementarity. Relative to the selected lexical channel, retrained low rank adds 0.73 unique evidence chunks per 2WikiMultiHopQA example and 0.77 per MuSiQue example. Yet the fused policy falls 0.054 below exact on 2WikiMultiHopQA, interval $[-0.100, -0.009]$, and 0.028 below validation-selected BM25 on MuSiQue, interval $[-0.047, -0.010]$. Twenty validation identities are not enough to calibrate a global mixture under a fixed budget:

$$\text{channel complementarity} \neq \text{reliable static fusion.} \tag{11}$$

A useful new routing channel expands the action space; it does not solve channel selection.

6 Compression Frontier: Quality versus Index Residency

Where the native channel helps, aggressive index compression preserves much of its value. On 2WikiMultiHopQA, rank-8/eight-centroid recall is 0.1763. Its gain over mean is 0.0786, interval $[0.0437, 0.1143]$, retaining 84.9% of the rank-16 gain while reducing FP32 routing-index state from 2,048 to 256 bytes/chunk, excluding masks. It is 16× smaller than one full-width mean key and 512× smaller than the full 32-token native-key stream. On MuSiQue the same retention ratio is not meaningful because neither low-rank gain is resolved.

The earlier QASPER frontier shows the complementary width result. Rank-16 all-token routing reaches 0.254 recall at 2,048 bytes/chunk, rank 8 reaches 0.246 at 1,024 bytes, and rank-8/eight-centroid reaches 0.183 at 256 bytes. Rank 32 falls to 0.218 despite more state and higher seed variance. More rank is not monotonically useful. This is consistent with the smaller projections acting partly as task-selective regularization: increasing rank may reintroduce native-response dimensions irrelevant to evidence ordering. The experiment does not isolate those dimensions, so this remains an interpretation rather than a causal claim.

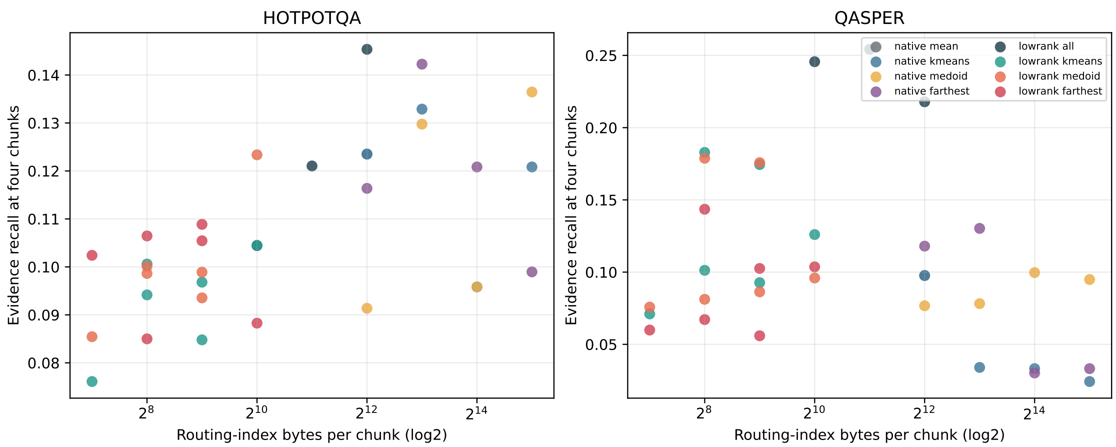

Figure 4: Evidence recall against persistent FP32 routing-index bytes per 32-token chunk. The plot excludes backing native K/V and selected-K/V transfer.

The reference production index stores FP32, FP16, BF16, or symmetric per-rank INT8 projected rows and performs batched stable top-k in one tensor path. Including masks, rank-16 FP16 uses 1,056 bytes/chunk, FP32 2,080, INT8 about 545, and INT8/eight-centroid about 137. On the GTX 950M, warm single-query search is 1.5–2.1 ms; quantization changes residency much more than latency at this cohort size. Four selected chunks transfer roughly 0.50 MB of unchanged native K/V in the fixed cross-dataset cost cohort, while backing K/V averages 7.15 MB for 2WikiMultiHopQA and 10.35 MB for MuSiQue. Index compression changes the address structure, not the backing-memory obligation.

7 Cross-Model Replication

SmolLM2-135M supplies a Llama-family geometry with a different tokenizer, 30 layers, nine query heads, three native K/V heads, and width-64 heads. At layer 29, rank-16 QASPER recall is 0.155 versus 0.092 for mean, a gain of 0.0638 with interval $[0.0192, 0.1148]$. Rank-8/eight-centroid reaches 0.165, gain 0.0739 with interval $[0.0226, 0.1342]$. Exact remains stronger at 0.298. On HotpotQA, neither learned index improves over mean and BM25 again leads.

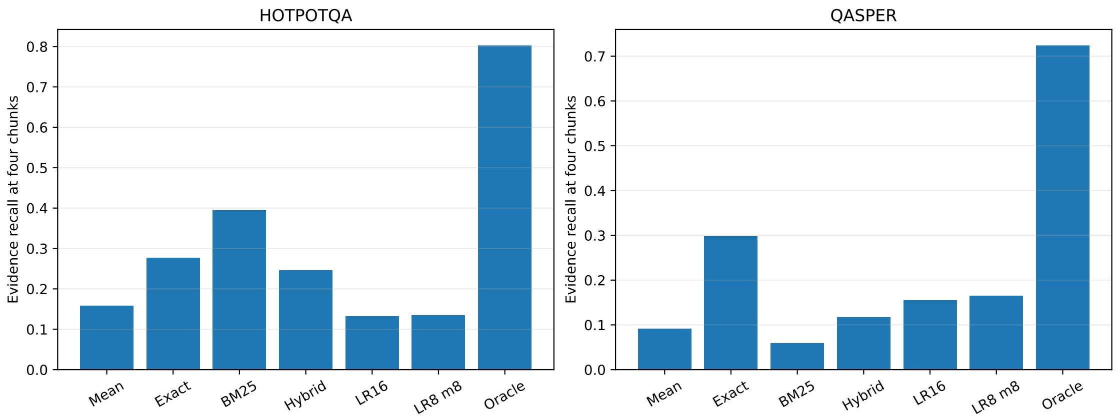

Figure 5: SmolLM2/Llama-family replication on the fresh Hotpot/QASPER cohort. The QASPER-over-mean effect and the lexical HotpotQA ordering both transfer.

This establishes QASPER portability across two model/tokenizer families. It does not establish a full model-by-dataset matrix: a SmolLM2 2WikiMultiHopQA feature capture would require a separate model-specific preprocessing run and is left for replication rather than added after the fact to this frozen study.

Table 4: What the portability evidence covers. "Not tested" is an unfilled replication cell, not a negative result.

Dataset	Qwen3-0.6B	SmolLM2-135M
QASPER	positive	positive
2WikiMultiHopQA	positive	not tested

8 Routing Is Not Reading

Saved chunk identities are replayed through unchanged layer-27 native K/V. Table 5 compares informed, shuffled, bottom-ranked, oracle, and direct-text conditions at the same four-chunk request budget. On QASPER, rank 16 exceeds shuffled memory by 0.3115 gold-token log-probability, interval $[0.1378, 0.4887]$, and bottom-ranked memory by 0.4699, interval $[0.2369, 0.7118]$. The selected identities therefore matter causally.

Yet rank 16 is 0.2124 below no memory, interval $[-0.4079, -0.0194]$. Even the evidence oracle is slightly negative, while direct evidence text is positive. The frozen late-layer consumer can distinguish better from worse memory without integrating selected native K/V beneficially. This is not a routing failure; it is a downstream consumption bottleneck and a direct motivation for PRA-aware training.

The experiment therefore separates four stages that are often conflated:

$$\boxed{\text{Discover}} \rightarrow \boxed{\text{Select}} \rightarrow \boxed{\text{Materialize}} \rightarrow \boxed{\text{Consume}}. \tag{12}$$

Paper 2.8 improves selection. It does not claim to have solved the frozen consumer's use of the materialized state.

Table 5: Mean gold-token log-probability change relative to no memory. All memory rows request four chunks.

Condition	HotpotQA	QASPER
Native mean	+0.087	-0.095
Rank 16	+0.074	-0.212
Rank 8, eight centroids	+0.034	-0.194
Shuffled chunks	+0.065	-0.524
Bottom-ranked chunks	-0.035	-0.682
Evidence oracle	+0.076	-0.092
Direct evidence text	+2.956	+0.225

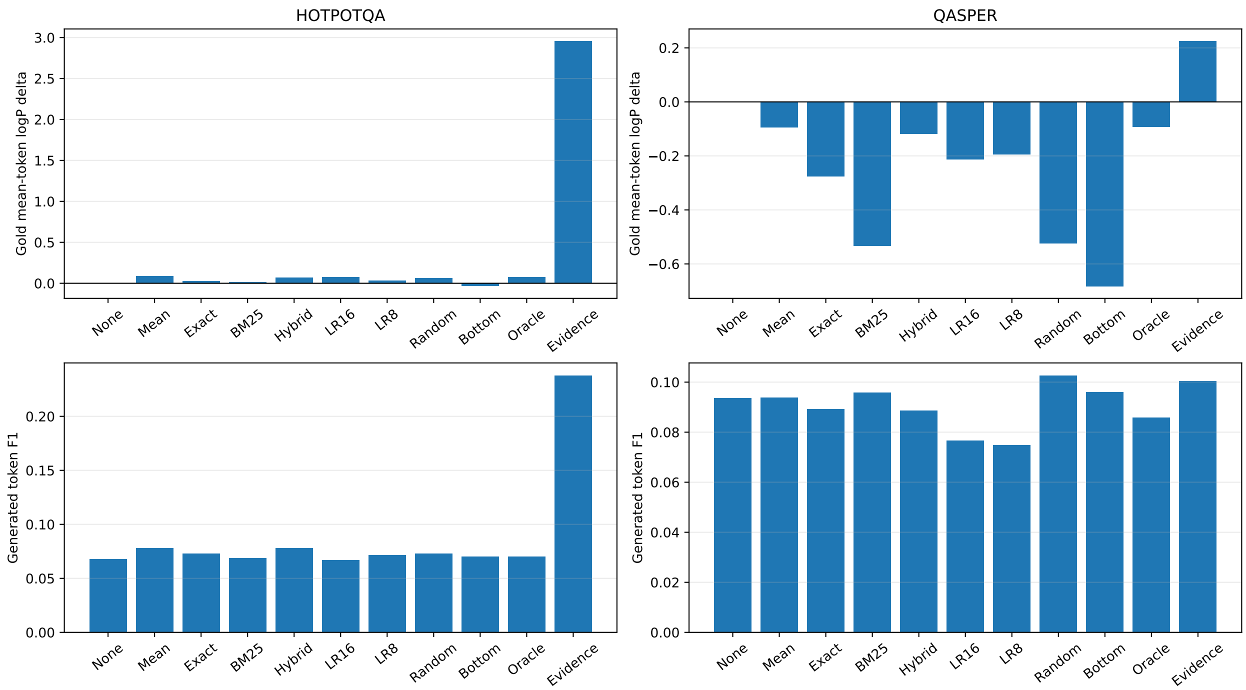

Figure 6: Native-K/V causal controls. Informed QASPER routing is better than wrong memory, but the frozen consumer still underperforms no memory.

9 Relation to Lexical, Semantic, and Associative Channels

The measured channel ordering is consistent with the broader PRA series. Paper 2.5 shows that native associative topology can solve controlled bridges without yielding a uniform natural frontier. Paper 2.6 establishes representation heterogeneity: exact and BM25 cues can dominate semantic summaries. Paper 2.7 shows that better query decomposition does not guarantee better retrieval. Paper 3.5 finds that adaptive channel/action selection remains difficult. Paper 2.8 adds native interaction features as another compact channel; it does not collapse these distinctions.

The emerging factorization is

$$\text{representation quality} \neq \text{retrieval utility} \neq \text{channel selection} \neq \text{memory consumption}. \tag{13}$$

QASPER and 2WikiMultiHopQA show that the low-rank channel is useful. Hotpot and MuSiQue show that lexical retrieval remains necessary. Static-fusion failure shows that channel diversity is not itself a selection policy. The native-K/V controls show that even correct selection does not train the consumer.

10 Related Work

10.1 Query-aware native-KV selection

Quest stores per-page key-coordinate bounds and uses the current query to load critical active-K/V pages [9]. RetrievalAttention builds attention-aware vector indexes over K/V rows held in CPU memory and retrieves a small active subset during decoding [10]. Both are close in their use of the model's query/key geometry. Their scope is active long-sequence attention. Paper 2.8 instead ranks logical long-memory chunk identities before PRA materialization, learns a low-rank interaction space, and leaves the selected backing K/V unchanged.

10.2 Low-rank attention and active-KV compression

Linformer historically approximates the sequence attention matrix with a low-rank projection to reduce attention complexity [11]. Eigen Attention performs attention in a derived low-rank subspace and stores a smaller active cache [12]. Palu caches low-rank intermediate states and reconstructs full keys and values during attention [13]. SALS combines latent-space K/V compression with sparse attention and selected reconstruction [14]. These methods compress or alter the state used by active attention. Paper 2.8 stores an auxiliary routing index and then activates original K/V; it makes no active-cache compression claim.

10.3 KV eviction, clustering, and streaming

H_2O retains recent and accumulated heavy-hitter tokens under an active-cache budget [15]. SnapKV uses an observation window to cluster and retain prompt positions important to each head [16]; RocketKV adds coarse eviction and fine sparse attention [18]. StreamingLLM keeps recent tokens plus attention sinks for stable bounded-memory streaming [17]. ClusterKV recalls semantically clustered active-cache tokens [19]. These methods manage which sequence K/V remains or is fetched for decoding. Paper 2.8 indexes an external logical memory namespace whose selected chunks remain recoverable in full.

10.4 Multi-vector retrieval

ColBERT independently encodes queries and documents and retains token-level late interaction rather than collapsing each document to one vector [20]. ColBERTv2 compresses that multi-vector corpus index with residual codes and denoised supervision [21]. The conceptual parallel is strong: multiple compact rows preserve sparse matching information erased by a mean. The difference is representation and consumer. ColBERT learns retrieval embeddings for document search; Paper 2.8 projects frozen LM-native query/key features to select memory that the same LM will later consume as native K/V.

10.5 Learned landmarks and compressed context

Landmark Attention trains special block landmarks to provide random access to long contexts [22]. Gist tokens compress prompts into reusable learned activations [23]; AutoCompressors recurrently produce summary vectors used as soft prompts [24]. These systems train in-model summary state or replace long context with compact representations. Paper 2.8's index is offline/cacheable, query-addressed, non-recurrent, and separate from the full-fidelity memory selected for reading.

10.6 Retrieval and long-term memory architectures

RETRO retrieves text chunks from a large corpus and consumes them through a trained encoder/cross-attention path [25]. Memorizing Transformers retrieve recent internal key/value pairs with approximate nearest neighbors [26]. Infini-attention combines local attention with bounded compressive memory [27], while Titans learns a test-time neural long-term memory module [28]. These works address how long-term information is represented and consumed. Paper 2.8 is narrower: it studies a compact address channel for an existing native-K/V memory substrate and empirically exposes consumption as a separate unsolved stage.

Table 6: Coarse taxonomy by addressed state and attention-visible outcome. Labels intentionally avoid implementation details that are not shared across members of a method family.

Method	Address or summary state	Online query role	Attention-visible outcome / scope
Paper 2.8	learned low-rank native-key rows	projected query–key scoring	selected original native K/V; external logical memory
Quest	per-page key bounds	query estimates critical pages	selected existing K/V pages; active sequence
RetrievalAttention	attention-aware CPU vector index	query retrieves relevant rows	retrieved original K/V rows; active sequence
Palu	low-rank cached intermediates	no separate retrieval stage	reconstructed keys/values; active cache
Eigen Attention	derived low-rank cache	attention in low-rank space	low-rank attention state; active cache
SALS	latent compressed cache	latent sparse token selection	sparse/reconstructed state; active cache
H_2O/SnapKV/RocketKV/ClusterKV	token, head, page, or cluster policy state	policy-dependent selection	retained or recalled K/V subset; active sequence
ColBERTv2	residual-coded token vectors	query–document late interaction	document retrieval result; corpus index
Landmark Attention	learned block landmarks	attention selects blocks	selected context blocks; model context
Gist/AutoCompressor	learned summary activations	no external retrieval stage	summary state stands in for context

11 Discussion

Response preservation is not evidence ranking. The full-key teacher is informative but not universal. The original selector can increase teacher top-four overlap while reducing evidence recall. Native attention keys encode syntax, discourse, generation state, and other functions besides document retrieval. The successful projection changes the objective from imitating every teacher preference to fitting an evidence-bearing interaction subspace. It is therefore a discriminative address projection, not a conventional compressor that succeeds by reconstructing the native response.

One mean erases useful native address information. Fresh QASPER and 2WikiMultiHopQA results establish this claim under matched chunk and materialization budgets. Multiple low-dimensional projected rows preserve a sparse interaction that one full-width average misses. The result is about the address surface, not reconstruction of the source or K/V.

Low-rank native interaction is a channel, not a universal retriever. Hotpot and MuSiQue remain lexical, and exact retrieval is stronger even on the positive 2WikiMultiHopQA cohort. The appropriate system is multi-representational: typed references, token-native exact/BM25, semantic/native interaction, and possibly associative traversal remain distinct controls.

Better routing is not enough. The causal replay establishes that selected identity matters while also showing that a frozen late-layer consumer can mishandle good memory. Paper 4's PRA-aware training should therefore retain shuffled, bottom-ranked, oracle, and no-memory controls. Otherwise consumption adaptation can be mistaken for routing progress or vice versa.

Compactness factorizes into width and multiplicity. Equation 9 explains why eight low-width representatives can occupy less index state than one full-width mean. The measured 4,096 $\rightarrow$ 256 FP32 routing-index bytes/chunk reduction, excluding masks, neither removes nor compresses the backing native K/V.

12 Limitations

The study uses one small Qwen model for all four datasets and one smaller SmolLM2 model for Hotpot/QASPER only. It tests one late routing layer, 32-token chunks, and a primary four-chunk budget. Five seeds quantify projection initialization, while identity bootstraps quantify cohort variation; neither establishes broad domain generalization. The 20-identity validation splits for 2WikiMultiHopQA and MuSiQue are small enough for task-specific projections and fusion weights to overfit. The study separately establishes cross-model transfer on QASPER and cross-dataset transfer from QASPER/Hotpot to 2WikiMultiHopQA; it does not establish their full Cartesian product. SmolLM2-by-2WikiMultiHopQA remains an explicitly untested replication cell.

Evidence supervision makes the projection task-adapted, not an unsupervised compressor. The index excludes backing native K/V, which still requires host or device storage and selected-state transfer. The generation replay uses one frozen late layer and a 0.6B model; it does not test trained multi-layer consumption. Component timings use eager PyTorch on an old mobile GPU. INT8 is dequantized rather than scored with a fused integer kernel, and deterministic centroid construction is an offline Python path. The controlled response suite contains only 20 schematic cases.

13 Conclusion

Representation. Low-rank native query–key projection preserves useful query-dependent address information that one mean key erases. The effect is reproducible on fresh QASPER, transfers to a second model family, and extends to 2WikiMultiHopQA, where a rank-8/eight-centroid index retains 84.9% of the rank-16 gain at approximately 256 FP32 routing-index bytes/chunk, excluding masks. The useful map is best understood as a discriminative projection for evidence ranking, not a faithful reconstruction of the full native response.

Boundary. The auxiliary index does not compress backing native K/V and is not a universal retriever. Hotpot and MuSiQue remain lexical, exact and BM25 remain necessary, static fusion fails, and better-selected native K/V can still harm a frozen consumer.

Next problem. The practical implication is to preserve low-rank native-QK as one compact address channel, while treating channel selection and memory consumption as independent problems for adaptive control and PRA-aware training.

A Full Response-Distillation Teacher Study

The initial program asked whether a small subset of actual native keys could preserve the full chunk response. For native query head h and token key i,

$$a_{i,h}(q, C) = \frac{q_h^\top k_{i,g(h)}}{\sqrt{d_h}}, \tag{14}$$

where grouped-query map $g(h)$ selects the corresponding native key head. A teacher reduces tokens per head and then averages query heads:

$$S^*(q, C) = \frac{1}{H_q} \sum_h F(a_{1,h}, \ldots, a_{T,h}). \tag{15}$$

Validation compares maximum, top-r mean, length-normalized log-sum-exp, and a softmax attention-mass proxy. Top-four mean is frozen before test because its validation evidence recall is 0.1387 versus 0.1255 for the matching mean-key control. Max, log-sum-exp, and attention mass do not beat that control.

On the controlled suite, full-key recall is 1.000 versus 0.950 for mean. On the original held-out natural cohort, full-key recall is 0.1240 versus 0.1235 on HotpotQA and 0.1252 versus 0.0976 on QASPER. The full response is therefore a weak natural evidence oracle: useful for mechanism diagnosis, but not a target that should be treated as evidence truth.

Preservation metrics include MAE, RMSE, Spearman ρ, Kendall τ, KL between softmax-normalized chunk distributions, and teacher top-k overlap for $k \in \{1, 2, 4, 8\}$. Their complete rows remain in the tracked artifacts.

B Landmark-Selector Baselines

All deployable landmark outputs are actual pretrained keys; mean is the sole synthetic baseline. Last and random retain fixed token subsets. Farthest-first greedily preserves cosine diversity. A non-deployable greedy oracle sees the evaluation query and adds the token that most reduces response error. The key-only learned selector scores eight features: key norm, mean head norm, head-norm variation, cosine to the chunk centroid, local key change, forward/reverse position, and validity. Its $8 \to 32 \to 1$ MLP has 321 parameters and receives no question, evidence label, dataset ID, or lexical feature at inference.

Table 7: Original held-out four-chunk routing. Learned rows average five seeds; oracle rows are non-deployable. Parentheses give teacher top-four overlap.

Method	HotpotQA	QASPER
Mean key, $m = 1$	0.124 (0.594)	0.098 (0.516)
Full-K teacher	0.124 (1.000)	0.125 (1.000)
Farthest-first, $m = 2$	0.142 (0.734)	0.130 (0.422)
Greedy oracle, $m = 4$	0.145 (0.922)	**0.178** (0.703)
Greedy oracle, $m = 8$	0.124 (0.984)	0.136 (0.812)
Learned selector, $m = 4$	0.122 (0.741)	0.083 (0.422)
Learned selector, $m = 8$	0.124 (0.834)	0.139 (0.553)

Greedy $m = 8$ improves aggregate teacher overlap by 0.3438, but $m = 4$ gives the larger QASPER recall gain. The learned $m = 8$ selector recovers only 40.5% of the greedy overlap gain, below the preregistered 80% threshold. QASPER seed recall ranges from 0.0478 to 0.1821 even though teacher overlap is more stable. The gate fails; synthetic off-manifold slots and recurrent memory remain closed.

C Query-Conditioned Controller Sweep

The continuation adds a bounded low-rank query interaction to token logits:

$$s_i = \mathrm{MLP}(f_i) + \frac{(W_q \bar{q})^\top (W_f f_i)}{\sqrt{r}}, \tag{16}$$

for $r \in \{8, 16, 32\}$ and $m \in \{4, 8\}$. Four validation-only objectives are crossed with rank, landmark count, and five seeds: oracle imitation, listwise evidence ranking, combined ranking plus response KL, and a decision-aware loss around the fourth-place boundary. The matrix contains 120 controllers.

The prespecified combined $r = 16, m = 4$ row reaches 0.1667 Hotpot recall but only 0.0574 QASPER recall. A test-selected exploratory decision-aware $r = 32, m = 8$ row reaches 0.1718 on QASPER, near exact at 0.1776, but its identity interval is wide and it cannot replace the primary result. Oracle-imitation $r = 32, m = 8$ has the best teacher overlap and recovers 47.3% of greedy-oracle gain, still below the deployment gate. Across configurations, teacher overlap correlates negatively with Hotpot recall and positively with QASPER recall. Response preservation, evidence ranking, and dataset utility are not interchangeable.

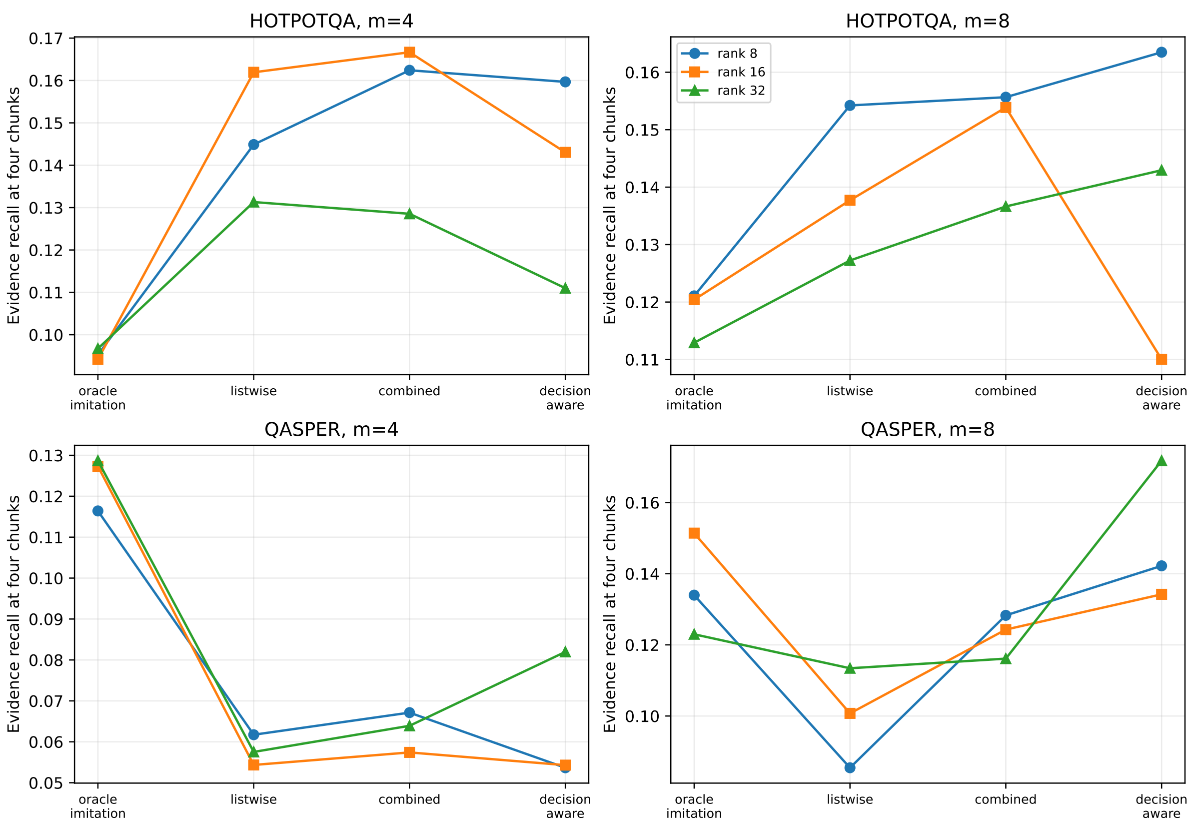

Figure 7: The 120-controller continuation. Objective changes both level and dataset direction; additional rank or landmarks are not monotonically useful.

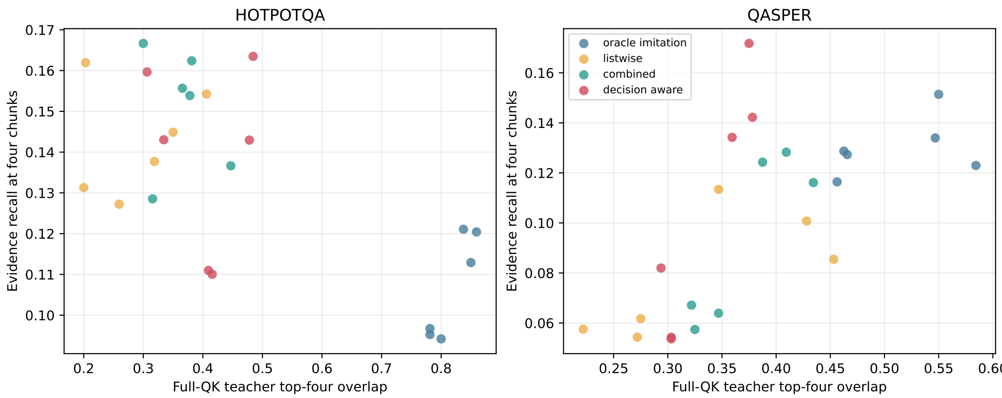

Figure 8: Teacher-response preservation versus evidence recall. The opposing dataset trends motivated the direct evidence-supervised low-rank formulation.

D Gate History

Table 8: Preregistered response-distillation gates.

Gate	Status	Evidence
G0	Pass	224 inherited Paper 2.6 rows reproduce with zero aggregate mismatches.
G1	Pass	Controlled full-K recall 1.00 vs. 0.95 mean; small QASPER teacher gain.
G2	Pass	Greedy $m = 8$ overlap +0.3438; greedy $m = 4$ QASPER recall +0.0802.
G3	Fail	Key-only recovers 40.5%; best query-conditioned imitation recovers 47.3%, below 80%.
G4	Closed	Synthetic slots require G3.
G5	Closed	Recurrent memory requires a successful offline compressor.

The post-G3 E-gates are diagnostic and do not revise G3. Frozen artifact parity passes; matched native geometry does not explain the query-aware effect; the prespecified controller is cross-dataset inconclusive; direct low-rank routing passes the quality/state frontier; and rank-8 static representatives open the cross-dataset study. The M-gates then pass feature/evidence audit, pass low-rank gain and compression on 2WikiMultiHopQA, fail static fusion, and establish usefulness of the unchanged architecture on QASPER plus 2WikiMultiHopQA. MuSiQue remains an explicit non-result.

E Controlled Synthetic and Native-Geometry Checks

Twenty synthetic cases cross chunk sizes 32, 64, 128, and 256 with 16 candidate chunks. Evidence chunks contain sparse query-aligned native keys while distractors have coherent weaker means. Full-key recall is 1.00 and mean recall 0.95. This verifies the sparse-address mechanism but is not natural-language evidence.

Deterministic Euclidean centroids, medoids, and farthest-first actual keys test whether ordinary within-chunk geometry explains the natural result. At matched $m = 2$, farthest-first is strongest at 0.142 Hotpot and 0.130 QASPER recall. Increasing to four or eight representatives is non-monotone; four native centroids fall to 0.033 QASPER recall despite 0.453 teacher overlap. Simple multimodal geometry does not recover the low-rank interaction result.

Structural selector ablations are also dataset-specific. Hotpot favors query-independent salience in one test-matrix row, while QASPER favors a bilinear-only variant. Their combination is not uniformly additive. These rows are exploratory diagnostics, not selected alternatives to the frozen rank-16 mechanism.

F Budget Sensitivity

The primary $K = 4$ contract is shared across methods. All new cross-dataset examples have two to four authored evidence items, so four chunks can touch each evidence region, though only 11.7% can contain every supporting token. At $K = 6$, 2WikiMultiHopQA exact recall is 0.330 and retrained rank 16 is 0.274; group completion is 0.26 and 0.34. At $K = 8$, MuSiQue BM25 reaches 0.183 recall and 0.36 completion versus 0.130 and 0.26 for retrained rank 16. These rows materialize roughly 191 and 255 tokens and do not change channel ordering.

G Production-Index Microbenchmarks

The vectorized index scores all chunks for a query batch, applies stable top-k, tracks masks and quantization scales, and appends chunks without rebuilding old rows. Eight-centroid clustering remains offline. On the GTX 950M, an eight-query batch sustains approximately 2,660–4,090 component queries/s. FP16 and INT8 preserve seed-11 FP32 selections in aggregate; centroid modes trade recall for residency. Append updates average below 0.21 ms except one warm-up-tainted row. These are routing-component measurements, not end-to-end serving throughput.

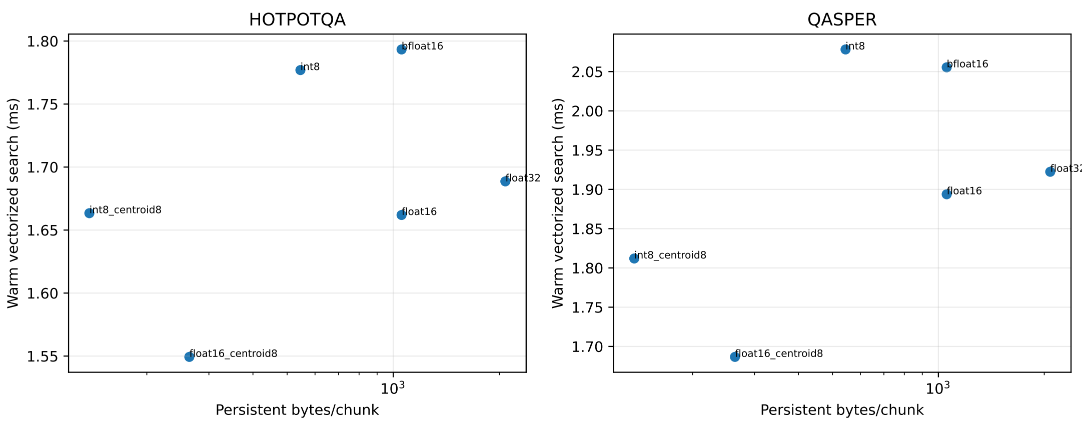

Figure 9: Warm vectorized routing against persistent index bytes. Quantization changes residency more than latency on this small GPU.

The separate 2WikiMultiHopQA/MuSiQue cost audit fixes seed 11 and measures 20 identities per dataset. Rank 16 averages 1.44/1.65 ms warm routing, native mean 1.92/2.62 ms, and exhaustive Python lexical scoring about 12.8/21.0 ms. Source encoding and native-K/V materialization are excluded. Rank-16 construction is 2.46/3.95 ms per example; centroid construction is 0.71/1.22 s and offline.

H Historical Matched Comparisons

On the original frozen Paper 2.6 identities and four-chunk budget, learned key-only QK routing is not a global frontier. Hotpot recall 0.124 is below BM25 0.398, exact 0.287, and iterative hybrid 0.279. QASPER recall 0.139 exceeds the old semantic gist 0.038 and iterative hybrid 0.081, but remains below exact 0.178. The later direct rank-16 projection replaces this key-only row in the main argument because it is both stronger and freshly confirmed.

Paper 2.5's native-QK closure used 256-token parents and a 20% final-parent budget. Its chain-completion rows concern associative successor discovery, not one-shot 32-token ranking, and are not pooled with this paper's recall tables.

I Reimplementation Map and Artifacts

The core serving path is:

```
q  = native_q_projection(frozen_query_state)      # [Hq, Dh]
K  = cached_pre_rope_keys                         # [C, T, Hkv, Dh]
Zk = cache(Wk @ flatten_native_keys(K))           # [C, T, r]
zq = Wq @ flatten_native_query(q)                 # [r]
s  = chunk_reduce(Zk @ zq / sqrt(r))              # [C]
ids = stable_topk(s, k=4)
materialize_native_kv(ids)                        # unchanged K/V
```

Static joint compression replaces each chunk's T projected rows with four or eight deterministic representatives. Only the address index changes.

- `src/pra_hf/qk_compression.py` defines native GQA mapping, teachers, selectors, low-rank routing, static representatives, metrics, and `LowRankRoutingIndex`.
- `run_gated_study.py` and `run_query_conditioned_study.py` reproduce the teacher/landmark program and 120-controller continuation.
- `run_low_rank_frontier.py` trains the direct projections; `run_selector_ablation.py` isolates salience and bilinear terms.
- `precompute_multidataset_native_qk.py` and `run_multidataset_extension.py` implement 2Wiki/MuSiQue capture, transfer, fusion, and M0–M5.
- `run_confirmation.py`, `run_confirmation_generation.py`, and `run_cross_model_replication.py` reproduce fresh routing, native-K/V causal controls, and SmolLM2.
- `run_production_index_benchmark.py` measures vectorized and quantized residency, search, and append operations.

Tracked row-level artifacts, manifests, bootstrap effects, checkpoints, and figures live under `docs/papers/shared/results/paper2_8_qk_compression/`. Large feature caches are reproducible from recorded source/model hashes and are intentionally not required for reading the result tables. The original test cache hash is `edc589cc...75993ae8`; validation is `87b85b25...7533cf8`.

References

[1] Simão, J. Progressive Retrieval Attention: Model-Bounded Sparse Native-KV for Long Context and URI-Addressed Memory. Project manuscript, 2026.

[2] Simão, J. Iterative PRA and Associative Memory Closure. Project manuscript, 2026.

[3] Simão, J. Hybrid Lexical-Semantic Routing for Progressive Retrieval Attention. Project manuscript, 2026.

[4] Simão, J. Latent Query Decomposition from Frozen Transformer State Graphs. Project manuscript, 2026.

[5] Yang, Z. et al. HotpotQA: A Dataset for Diverse, Explainable Multi-hop Question Answering. EMNLP, 2018.

[6] Dasigi, P. et al. A Dataset of Information-Seeking Questions and Answers Anchored in Research Papers. NAACL, 2021.

[7] Ho, X., Nguyen, A.-K. D., Sugawara, S., and Aizawa, A. Constructing a Multi-hop QA Dataset for Comprehensive Evaluation of Reasoning Steps. COLING, 2020.

[8] Trivedi, H. et al. MuSiQue: Multihop Questions via Single-hop Question Composition. TACL 10, 2022.

[9] Tang, J., Zhao, Y., Zhu, K., Xiao, G., Kasikci, B., and Han, S. Quest: Query-Aware Sparsity for Efficient Long-Context LLM Inference. ICML, 2024.

[10] Liu, D. et al. RetrievalAttention: Accelerating Long-Context LLM Inference via Vector Retrieval. arXiv:2409.10516, 2024.

[11] Wang, S., Li, B. Z., Khabsa, M., Fang, H., and Ma, H. Linformer: Self-Attention with Linear Complexity. arXiv:2006.04768, 2020.

[12] Saxena, U., Saha, G., Choudhary, S., and Roy, K. Eigen Attention: Attention in Low-Rank Space for KV Cache Compression. Findings of EMNLP, 2024.

[13] Chang, C.-C. et al. Palu: KV-Cache Compression with Low-Rank Projection. ICLR, 2025.

[14] Zhang, Y. et al. SALS: Sparse Attention in Latent Space for KV Cache Compression. NeurIPS, 2025.

[15] Zhang, Z. et al. H_2O: Heavy-Hitter Oracle for Efficient Generative Inference of Large Language Models. NeurIPS, 2023.

[16] Li, Y. et al. SnapKV: LLM Knows What You Are Looking for before Generation. NeurIPS, 2024.

[17] Xiao, G., Tian, Y., Chen, B., Han, S., and Lewis, M. Efficient Streaming Language Models with Attention Sinks. ICLR, 2024.

[18] Behnam, P. et al. RocketKV: Accelerating Long-Context LLM Inference via Two-Stage KV Cache Compression. arXiv:2502.14051, 2025.

[19] Liu, G., Li, C., Zhao, J., Zhang, C., and Guo, M. ClusterKV: Manipulating LLM KV Cache in Semantic Space for Recallable Compression. arXiv:2412.03213, 2024.

[20] Khattab, O. and Zaharia, M. ColBERT: Efficient and Effective Passage Search via Contextualized Late Interaction over BERT. SIGIR, 2020.

[21] Santhanam, K., Khattab, O., Saad-Falcon, J., Potts, C., and Zaharia, M. ColBERTv2: Effective and Efficient Retrieval via Lightweight Late Interaction. NAACL, 2022.

[22] Mohtashami, A. and Jaggi, M. Landmark Attention: Random-Access Infinite Context Length for Transformers. NeurIPS, 2023.

[23] Mu, J., Li, X. L., and Goodman, N. Learning to Compress Prompts with Gist Tokens. NeurIPS, 2023.

[24] Chevalier, A., Wettig, A., Ajith, A., and Chen, D. Adapting Language Models to Compress Contexts. EMNLP, 2023.

[25] Borgeaud, S. et al. Improving Language Models by Retrieving from Trillions of Tokens. ICML, 2022.

[26] Wu, Y., Rabe, M. N., Hutchins, D., and Szegedy, C. Memorizing Transformers. ICLR, 2022.

[27] Munkhdalai, T., Faruqui, M., and Gopal, S. Leave No Context Behind: Efficient Infinite Context Transformers with Infini-attention. arXiv:2404.07143, 2024.

[28] Behrouz, A., Zhong, P., and Mirrokni, V. Titans: Learning to Memorize at Test Time. arXiv:2501.00663, 2025.